

HMMT November General — Q6/10

Jonathan Kasongo

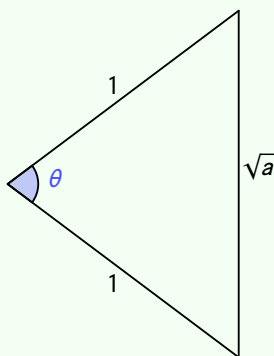
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Call a polygon *normal* if it can be inscribed within a unit circle. How many non-congruent normal polygons are there such that the square of each side length is a positive integer?

Solution

The idea is that we can dissect any such polygon into a finite number of triangles that look like so,



where a is a positive integer. Now from Cosine law,

$$a = 2(1 - \cos \theta)$$

$$\cos \theta = 1 - \frac{a}{2}$$

Since $\cos \theta \in [-1, 1]$ and $a \in \mathbb{N}$, we can check that $a \in \{1, 2, 3, 4\}$. Note that $\theta \in [0, 180]$ since the angle is in a triangle. Now plugging in each value of a and solving yields

$$\theta \in \{60, 90, 120, 180\}.$$

Since any normal polygon is just a collection of these triangles, we are just looking for solutions to

$$60a + 90b + 120c + 180d = 360$$

$$2a + 3b + 4c + 6d = 12.$$